

Module-7

Section A

1) Construct a forward difference table for the data in given table:

x	10	11	12	13
y	22	24	28	34

$\Rightarrow$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
10	22			
11	24	2		
12	28	4	2	
13	34	6	2	0

2) Write down the Newton's backward interpolation formula:

$$\Rightarrow y(x) = y_0 + p \nabla y_0 + \frac{p(p+1)}{2!} \nabla^2 y_0 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_0 \text{ where } p = \frac{x - x_n}{h}$$

3) Write the Newton's general interpolation formula up to fourth divided differences?

$$\Rightarrow y = f(x) = f(x_0) + (x - x_0) f(x_0, x_1) + (x - x_0)(x - x_1) f(x_0, x_1, x_2) + (x - x_0)(x - x_1)(x - x_2) f(x_0, x_1, x_2, x_3) + (x - x_0)(x - x_1)(x - x_2)(x - x_3) f(x_0, x_1, x_2, x_3, x_4)$$

4) Write down the formula to find a real root of an equation $f(x) = 0$ using Regula-Falsi method?

Formula for Regula - falsi method

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

5) Find the initial approximation of the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 1$ using modified Euler's method taking $h = 0.05$?

$$\Rightarrow x_0 = 0, y_0 = 1, h = 0.05, f(x, y) = x + y$$

$$f(x_0, y_0) = x_0 + y_0 = 0 + 1 = 1$$

By Euler's formula

$$y_1^{(0)} = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.05 \times 1 = 1.05$$

$$y_1^{(0)} = 1.05$$

Section B.

Using Lagrange's interpolation formula, find the value of y for $x = 10$ given.

x	5	6	9	11
y	12	13	14	16

$$\Rightarrow \left. \begin{array}{llll} x_0 = 5 & x_1 = 6 & x_2 = 9 & x_3 = 11 \\ y_0 = 12 & y_1 = 13 & y_2 = 14 & y_3 = 16 \end{array} \right\} \begin{array}{l} x = 10 \\ y = ? \end{array}$$

We have Lagrange's interpolation formula.

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)y_0}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} + \frac{(x-x_0)(x-x_2)(x-x_3)y_1}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \\ + \frac{(x-x_0)(x-x_1)(x-x_3)y_2}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} + \frac{(x-x_0)(x-x_1)(x-x_2)y_3}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

$$\Rightarrow \frac{4 \times 1 \times (-1) \times 12}{(-1)(-4)(-6)} + \frac{5 \times 1 \times (-1) \times 13}{1 \times (-3)(-5)} + \frac{(5)(4)(-1)14}{(4)(3)(-2)} + \frac{5 \times 4 \times 1 \times 16}{6 \times 5 \times 2}$$

$$= 2 - \frac{13}{3} + \frac{35}{3} + \frac{16}{3} = \frac{44}{3} = 14.6667$$

$$y(10) = 14.6667$$

2) Employ Taylor's series method to find y at $x = 0.1$ correct to 4 decimal places considering up to fourth degree term given the equation.

$$\frac{dy}{dx} = 2y + 3e^x, y(0) = 0$$

$\Rightarrow$ By data, $y' = 2y + 3e^x$ and $y_0 = 0$ i.e., $x_0 = 0, y_0 = 0$
Taylor's series expansion is given by

$$y(x) = y(x_0) + (x-x_0)y'(x_0) + \frac{(x-x_0)^2}{2!}y''(x_0) + \frac{(x-x_0)^3}{3!}y'''(x_0) + \frac{(x-x_0)^4}{4!}y^{(4)}(x_0) \rightarrow (1)$$

$\therefore$ we shall compute $y(0.1)$

$$\text{consider } y' = 2y + 3e^x; y'(0) = 3$$

$$y'' = 2y' + 3e^x; y''(0) = 2 \times 3 + 3 = 9$$

$$y''' = 2y'' + 3e^x; y'''(0) = 2 \times 9 + 3 = 21$$

$$y^{(4)} = 2y''' + 3e^x; y^{(4)}(0) = 2 \times 21 + 3 = 45$$

From (1) we have

$$y(0.1) = y(0) + 0.1 y'(0) + \frac{(0.1)^2}{2} y''(0) + \frac{(0.1)^3}{6} y'''(0) + \frac{(0.1)^4}{24} y^{(4)}(0)$$

$$= 0 + 0.1 \times 3 + \frac{0.01}{2} \times 9 + \frac{0.001}{6} \times 21 + \frac{0.0001}{24} \times 45$$

$$y(0.1) = 0.3487$$

3) Use Newton-Raphson method to find a root correct to three decimal of the equation $3x = \cos x + \tan x = 0.5$
 $\Rightarrow$ given $f(x) = 3x - \cos x - 1$ and $x_0 = 0.5$

$$f'(x) = 3 + \sin x$$

By Newton-Raphson method, we have.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{where } n=0, 1, 2, \dots$$

$$= x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{(3x_n - \cos x_n - 1)}{3 + \sin x_n} = \frac{x_n \sin x_n + \cos x_n + 1}{3 + \sin x_n} \quad \text{--- (1)}$$

This is the required iterative formula.

First approximation.

$$\text{For } n=0, x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{(-0.378)}{3.479}$$

$$x_1 = 0.609$$

2nd approximation

$$\text{For } n=1, x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.609 - \frac{f(0.609)}{f'(0.609)} = 0.609 - \frac{0.007}{3.572}$$

$$x_2 = 0.607$$

3rd approximation.

$$\text{For } n=2, x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.607 - \frac{f(0.607)}{f'(0.607)} = 0.607 - \frac{(-0.0003)}{3.570}$$

$$x_3 = 0.607$$

The approximate value of the root correct to 3 decimal places is 0.607.

4) use width's rule to evaluate $\int_0^1 \frac{2dx}{1+x^2}$ taking seven ordinates and hence find $\log_e 2$?

$\Rightarrow$ we divide the interval $[0, 1]$ into 6 equal

parts
 $(n+1)$ ordinates = n equal parts

$$n=6, h = \frac{1}{6} = \frac{b-a}{n} = \frac{1-0}{6} = \frac{1}{6}$$

The values of x and $y = \frac{1}{1+x}$ are calculated below

$$x \quad y = \frac{1}{1+x}$$

$$a = x_0 = 0$$

$$y_0 = 1$$

$$x_1 = 1/6$$

$$y_1 = 0.8571$$

$$x_2 = 2/6$$

$$y_2 = 0.7500$$

$$x_3 = 3/6$$

$$y_3 = 0.6667$$

$$x_4 = 4/6$$

$$y_4 = 0.6000$$

$$x_5 = 5/6$$

$$y_5 = 0.5555$$

$$b = x_6 = 1$$

$$y_6 = 0.5000$$

$$I = \int_a^b y dx = \frac{3h}{8} (y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})$$

for $n=6$

$$I = \int_a^b y dx = \frac{3h}{8} (y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3$$

$$I = 3(0.1667)(1 + 0.5000) + 3(0.8571 + 0.7500 + 0.6000 + 0.5555) + 2(0.6667)$$

for finding $\log_e$ (natural logarithm) first we have to integrate given integral

$$\int_0^1 \frac{dx}{1+x} = 0.6932$$

$$\frac{1}{1} [\log_e(1+x)]_0^1 \Rightarrow \frac{1}{1} [\log_e^2 - \log_e^1]$$

$$\frac{1}{1} [\log_e(\frac{2}{1})] = 0.6932 \Rightarrow \frac{1}{1} \times \log_e^2 = 0.6932$$

$$\log_e^2 = 0.6932$$

Section - C

1) Evaluate $f(9)$ and $f(18)$ using Newton's divided difference formula, given:

x	5	7	11	13	17
$f(x)$	150	392	1452	2366	5202

$\Rightarrow$ The divided difference table is

x	$f(x)$	1 st D.D	2 nd D.D	3 rd D.D	4 th D.D
$x_0 = 5$	$f(x_0) = 150$	$f(x_0, x_1)$ $\frac{392 - 150}{7 - 5} = 121$	$f(x_0, x_1, x_2)$ $\frac{265 - 121}{11 - 5} = 24$	$f(x_0, x_1, x_2, x_3)$ $\frac{32 - 24}{13 - 5} = 1$	$f(x_0, x_1, x_2, x_3, x_4)$ $\frac{42 - 43}{17 - 7} = -1$
$x_1 = 7$	$f(x_1) = 392$				
$x_2 = 11$	$f(x_2) = 1452$	$f(x_1, x_2)$ $\frac{1452 - 392}{11 - 7} = 265$	$f(x_1, x_2, x_3)$ $\frac{457 - 265}{13 - 11} = 32$	$f(x_1, x_2, x_3, x_4)$ $\frac{42 - 43}{17 - 11} = -1$	$f(x_1, x_2, x_3, x_4, x_5)$ $\frac{42 - 43}{17 - 11} = -1$
$x_3 = 13$	$f(x_3) = 2366$	$f(x_2, x_3)$ $\frac{2366 - 1452}{13 - 11} = 457$			
$x_4 = 17$	$f(x_4) = 5202$	$f(x_3, x_4)$ $\frac{5202 - 2366}{17 - 13} = 709$	$f(x_2, x_3, x_4)$ $\frac{709 - 457}{17 - 13} = 42$		

We have Newton's divided difference formula,

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + (x - x_0)(x - x_1)(x - x_2)f(x_0, x_1, x_2, x_3) + \dots$$

$$f(9) = 150 + (9 - 5)121 + (9 - 5)(9 - 7)24 + (9 - 5)(9 - 7)(9 - 11)1 + 0 = 810$$

$$f(18) = 150 + (18 - 5)121 + (18 - 5)(18 - 7)24 + (18 - 5)(18 - 7)(18 - 11)1 + 0 = 6158$$

2) From the data given below, find the number of students who obtained (i) less than 45 marks and (ii) between 40 and 45 marks?

x	Marks	30-40	40-50	50-60	60-70	70-80
$f(x)$	No. of Students	31	42	51	35	31

$\Rightarrow$ we shall substitute the given table with $f(x)$ representing the number of students less than x marks.
That is,

less than 40 marks 31 Students
 less than 50 marks $31+42 = 73$ Students
 less than 60 marks $73+51 = 124$ Students
 less than 70 marks $124+35 = 159$ Students
 less than 80 marks $159+31 = 190$ Students

This problem can be solved by Newton's forward interpolation formula.

The forward difference table is

x	$f(x) = y$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
40	31	$\Delta y_0 = 42$	$\Delta^2 y_0 = 9$	$\Delta^3 y_0 = -25$	$\Delta^4 y_0 = 37$
50	73	51	-16	12	
60	124	35	-4		
70	159	31			
80	190				

we shall find $f(45)$ by applying Newton's forward interpolation formula.

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots$$

$$\text{where } p = \frac{x - x_0}{h} = \frac{45 - 40}{10} = 0.5$$

$$f(45) = y(45) = 31 + 0.5 \times 42 + \frac{(0.5)(-0.5)}{2} + \frac{0.5 \times (-0.5)(-1.5)}{6} + \frac{0.5 \times (-0.5) \times (-1.5) \times (-2.5)}{24} \quad (37)$$

$$y(45) = 47.86 \approx 48$$

Thus the number of students obtaining less than 45 marks is 48

ii) we need to find $f(45) - f(40)$

$$\text{we know that } f(45) = 48, f(40) = 31 \\ = 48 - 31 = 17$$

Thus the number of students scoring marks between 40 and 45 is 17.

3) Apply Runge-Kutta method of fourth order to find an approximate value of y at $x = 0.1$ given that

$$\frac{dy}{dx} = x + y^2 \text{ and } y = 1 \text{ when } x = 0$$

$\Rightarrow$ By data $f(x, y) = x + y^2$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$

We shall first compute k_1, k_2, k_3, k_4

$$k_1 = h f(x_0, y_0) = 0.1 \times f(0, 1) = 0.1 \times (0 + 1^2) = 0.1$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.1 \times f(0.05, 1.05)$$

$$= 0.1 \times (0.05 + (1.05)^2) = 0.1 \times 1.1525 = 0.11525$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1 \times f(0.05, 1.057625)$$

$$= 0.1 \times (0.05 + (1.057625)^2)$$

$$= 0.1 \times 1.1686 = 0.11686$$

$$k_4 = h f(x_0 + h, y_0 + k_3) =$$

$$= 0.1 \times f(0.1, 1.11686)$$

$$= 0.1 \times [0.1 + (1.11686)^2]$$

$$= 0.1 \times 1.3474 = 0.13474$$

Formula for Runge-Kutta method of fourth order is given by

$$y(x_0 + h) = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

$$y(0.1) = 1 + \frac{1}{6} [0.1 \times 2(0.11525) + 2(0.11686) + 0.13474]$$

$$= 1 + \frac{1}{6} (0.69896) = 1 + 0.1165 = 1.1165$$

$$\therefore y(0.1) = 1.1165,$$

Module: 2

Section A

1) From a partial differential Equation from

$$z = (x+a)(y+b)?$$

=> By data, $z = (x+a)(y+b)$

Differentiating partially w.r.t x and y ,

$$P = \frac{\partial z}{\partial x} = (y+b)$$

$$Q = \frac{\partial z}{\partial y} = (x+a)$$

using (2) and (3) in (1) we obtain $z = PQ$

Thus $z = PQ$ is the required PDE.

2) From a partial differential Equation from

$$z = f(x+ay)?$$

=> Given $z = f(x+ay) \rightarrow (1)$

diff (1) partially w.r.t x

$$\frac{\partial z}{\partial x} = f'(x+ay) \rightarrow (2)$$

diff (1) partially w.r.t y we get

$$\frac{\partial z}{\partial y} = a f'(x+ay) \rightarrow (3)$$

Substituting (2) in (3) we get

$$\boxed{\frac{\partial z}{\partial y} = a \frac{\partial z}{\partial x}}$$

- 3) write down the Partial differential Equation for
 (i) one dimensional wave Equation
 (ii) one dimensional heat Equation.

$$\Rightarrow \text{i) } \frac{\partial^2 u}{\partial t^2} = c \frac{\partial^2 u}{\partial x^2}$$

$$\text{ii) } \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

- 4) write down the auxiliary Equation for the
 Lagrange's P.D.E; $(y+z)P + x(yQ - z) = 0$

> Given $(y+z)P + x(yQ - z) = 0$

$$(y+z)P + x y Q = x z$$

$$\therefore P = y+z, \quad Q = xy, \quad R = xz$$

$\therefore$ the auxiliary Equation are

$$\frac{\partial x}{y+z} = \frac{\partial y}{xy} = \frac{\partial z}{xz}$$

- 5) Form a partial differential Equation from
 $z = ax + by + ab$

$$\Rightarrow z = ax + by + ab \quad \text{--- (1)}$$

Partially differentiate Eqn (1) w.r.t (x and y)

$$\frac{\partial z}{\partial x} = P = a \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial y} = Q = b \quad \text{--- (3)}$$

Substitute 2 and 3 in 1

$$z = Px + Qy + Py$$

$$\therefore z = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + \left[\frac{\partial z}{\partial x} \right] \left[\frac{\partial x}{\partial y} \right]$$

Section B

7. Form a partial differential equation by eliminating arbitrary constants from

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$\Rightarrow$ given

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{--- (1)}$$

diff w.r.t (x and y)

$$\frac{\partial x}{\partial a^2} + \frac{\partial z}{\partial c^2} \cdot \frac{\partial z}{\partial x} = 0 \Rightarrow \frac{x}{a^2} + \frac{z}{c^2} \cdot \frac{\partial z}{\partial x} = 0 \quad \text{--- (2)}$$

$$\frac{\partial y}{\partial b^2} + \frac{\partial z}{\partial c^2} \cdot \frac{\partial z}{\partial y} = 0 \Rightarrow \frac{y}{b^2} + \frac{z}{c^2} \cdot \frac{\partial z}{\partial y} = 0 \quad \text{--- (3)}$$

differentiate 2 w.r.t (x)

$$\frac{1}{a^2} + \frac{1}{c^2} \left(z \frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial x} \right) = 0 \Rightarrow \frac{1}{a^2} + \frac{1}{c^2} (z x + p^2) = 0 \quad \text{--- (4)}$$

diff 3

from eqⁿ 2, we have

$$\frac{x}{a^2} = -\frac{z}{c^2} \cdot \frac{\partial z}{\partial x}$$

$$\frac{1}{a^2} = -\frac{z}{c^2} \cdot \frac{\partial z}{\partial x}$$

from eqⁿ (4), we have,

$$\frac{1}{a^2} = -\frac{1}{c^2} (z x + p^2)$$

$$-\frac{z}{c^2} \cdot \frac{\partial z}{\partial x} = -\frac{1}{c^2} (z x + p^2)$$

$$\frac{\partial z}{\partial x} = x (z x + p^2)$$

$$z = \frac{\partial z}{\partial x} = x x z \frac{\partial^2 z}{\partial x^2} + x \left(\frac{\partial z}{\partial x} \right)^2$$

2) Solve $y^2 z p = x^2 (2q + y)$?

$\Rightarrow$ The given equation $y^2 z p - x^2 z q = x^2 y$ is of the form $Pp + Qq = R$. The auxiliary equations are,

$$\frac{dx}{y^2 z} = \frac{dy}{-x^2 z} = \frac{dz}{x^2 y}$$

Taking the first and second terms we have
 $x^2 dx + y^2 dy = 0$

Integrating we get $x^3/3 + y^3/3 = c_1$ or $x^3 + y^3 = 3c_1$

Also taking the last two terms in (1) we have
 $y dy + z dz = 0$. Integrating we get

$$\frac{y^2}{2} + \frac{z^2}{2} = c_2 \text{ or } y^2 + z^2 = 2c_2$$

Thus a general solution of the PDE is given by

$$f(x^3 + y^3 ; y^2 + z^2) = 0$$

3) Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$?

The given PDE can be written in the form

$$\frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} \left[\frac{\partial z}{\partial y} \right] \right] = \cos(2x + 3y)$$

Integrate w.r.t x and treating y as a constant.

$$\frac{\partial}{\partial x} \left[\frac{\partial z}{\partial y} \right] = \int \cos(2x + 3y) dx$$

$$\frac{\partial}{\partial x} \left[\frac{\partial z}{\partial y} \right] = \frac{\sin(2x + 3y)}{2} + f(y)$$

Integrate w.r.t x and treating y as a constant

$$\frac{\partial z}{\partial y} = \frac{1}{2} \int \sin(2x + 3y) dx + f(y) \int (1 dx + g(y))$$

$$\frac{\partial z}{\partial y} = -\cos\left(\frac{2x+3y}{4}\right) + x f(y) + g(y)$$

Integrate w.r.t y and treating x as a constant.

$$z = -\frac{1}{4} \int \cos\left(\frac{2x+3y}{4}\right) + x \int f(y) dy + \int g(y) + h(x)$$

$$z = -\frac{\sin\left(\frac{2x+3y}{4}\right)}{\frac{3}{4}} + x f(y) + \eta(y) + h(x)$$

$$\text{where } f(y) = \int f(y)$$

$$\eta(y) = \int g(y)$$

4) Form a partial differential equation by eliminating the arbitrary functions f and g from $z = f(x+ay) + g(x-ay)$?

⇒ Given

$$z = f(x+ay) + g(x-ay) \rightarrow (1)$$

diff it partially twice w.r.t x and y

$$p = \frac{\partial z}{\partial x} = f'(x+ay) + g'(x-ay) \rightarrow (2)$$

$$q = \frac{\partial z}{\partial y} = a f'(x+ay) + a g'(x-ay) \rightarrow (3)$$

$$r = \frac{\partial^2 z}{\partial x^2} = f''(x+ay) + g''(x-ay) \rightarrow (4)$$

$$t = \frac{\partial^2 z}{\partial y^2} = a^2 f''(x+ay) + a^2 g''(x-ay) \rightarrow (5)$$

from eqn (5), we have

$$t = a^2 [f''(x+ay) + g''(x-ay)] \rightarrow (6)$$

Substitute eqn (2) in the eqn (6)

$$t = a^2 r$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

Question 1:

1) Form a partial differential Equation from.

$$\phi(x+y+z, x^2+y^2+z^2)=0?$$

$\Rightarrow$ We have by data

$$\phi(u, v) = 0 \quad \text{--- (1)}$$

where $u = (x+y+z)$ and $v = (x^2+y^2+z^2)$

differentiate u and v w.r.t x and y

$$\frac{\partial u}{\partial x} = 1 + \frac{\partial z}{\partial x} = 1+p \quad \text{--- (2)} \quad \frac{\partial v}{\partial x} = 2x + 2z \frac{\partial z}{\partial x} = 2(x+zp) \quad \text{--- (4)}$$

$$\frac{\partial u}{\partial y} = 1 + \frac{\partial z}{\partial y} = 1+q \quad \text{--- (3)} \quad \frac{\partial v}{\partial y} = 2y + 2z \frac{\partial z}{\partial y} = 2(y+zc) \quad \text{--- (5)}$$

Let us differentiate 1 w.r.t x and y . by applying chain rule.

$$\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial x} = 0$$

$$\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial \phi}{\partial u} \frac{\partial u}{\partial x} = - \frac{\partial \phi}{\partial v} \frac{\partial v}{\partial x} \quad \text{--- (6)}$$

$$\frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow \frac{\partial \phi}{\partial u} \cdot \frac{\partial u}{\partial y} = - \frac{\partial \phi}{\partial v} \cdot \frac{\partial v}{\partial y} \quad \text{--- (7)}$$

Dividing 6 by 7 we obtaine

$$\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = \frac{\frac{\partial v}{\partial x}}{\frac{\partial v}{\partial y}}$$

from Eq's 1, 2, 3 and 5

$$\frac{1+p}{1+q} = \frac{x(x+zp)}{x(y+za)}$$

$$(1+p)(y+za) = (1+q)(x+zp)$$

$$y+za+py + pqz = x+zp+qx+2pa$$

$$py - pz + qz - qx = x - y$$

$$p(y-z) + q(z-x) = x-y$$

$$\text{or, } (y-z) \frac{\partial z}{\partial x} + (z-x) \frac{\partial z}{\partial y} = x-y$$

Two P's the required P.D.E.

2) Solve $\frac{\partial^2 z}{\partial y^2} = z$, given that when $y=0$, $z=e^x$ and

$$\frac{\partial z}{\partial y} = e^{-x} ?$$

$\Rightarrow$ Let us assume that z is a function of y only
The given PDE assumes the form of ODE.

$$\frac{d^2 z}{dy^2} = z \quad \text{or} \quad \frac{d^2 z}{dy^2} - z = 0$$

$$(D^2 - 1)z = 0 \quad \text{where} \quad D = \frac{\partial}{\partial y} \quad D^2 = \frac{\partial^2}{\partial y^2}$$

$$\text{Auxiliary Equation } m^2 - 1 = 0$$

$$m = \pm 1$$

The roots are real and distinct the solution of ODE is given by.

$$z = c_1 e^y + c_2 e^{-y}$$

Solution of the PDE is got replacing c_1 and c_2 by functions of x .

The solution of the PDE is given by

$$z = f(x)e^y + g(x)e^{-y} \quad \text{--- (1)}$$

Now, we shall apply the given condition to
 find $f(x)$ and $g(x)$.
 using the condition when $y=0$, $z=e^x$ in (1), we have,
 after $e^x = f(x) + g(x) \rightarrow (2)$

differentiate w.r.t y ,

$$\frac{\partial z}{\partial y} = f(x)e^y - g(x)e^{-y}$$

using the condition when $y=0$, $\frac{\partial z}{\partial y} = e^{-x}$ in eqn (2)
 $e^{-x} = f(x) - g(x) \rightarrow (3)$

By solving the eqns (2) and (3)

$$\begin{aligned} f(x) + g(x) &= e^x \\ f(x) - g(x) &= e^{-x} \end{aligned}$$

$$\begin{aligned} 2f(x) &= e^x + e^{-x} \\ f(x) &= \frac{e^x + e^{-x}}{2} = \cosh x \end{aligned}$$

$$\begin{aligned} f(x) + g(x) &= e^x \\ f(x) - g(x) &= e^{-x} \end{aligned}$$

$$\begin{aligned} \frac{+}{-} \quad \frac{+}{-} \\ \hline 2g(x) &= e^x - e^{-x} \\ g(x) &= \frac{e^x - e^{-x}}{2} = \sinh x \end{aligned}$$

then by substituting $f(x)$ and $g(x)$ in

eqn (1)

$$z = \cosh x e^y + \sinh x e^{-y}$$

$\therefore$ is the required solution,

3) Obtain the various possible solutions of one dimensional heat equation by method of separation of variables?

$\Rightarrow$

$$\Rightarrow u_t = c^2 u_{xx}$$

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} \right)$$

heat Equation,

$$\text{consider } \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Let $u = XT$ where $x = x(x)$ and $T(t)$

here $u = XT$ be the solution

then the PDE becomes

$$x \frac{\partial T}{\partial t} = c^2 T \frac{\partial^2 x}{\partial x^2}$$

$$x \frac{\partial T}{\partial t} = c^2 T \frac{\partial^2 x}{\partial x^2}$$

dividing the above Equation by $c^2 XT$

$$\frac{1}{c^2 T} \frac{dT}{dt} = \frac{1}{x} \frac{d^2 x}{dx^2}$$

Equating both sides to a common constant k we have.

$$\frac{1}{c^2 T} \frac{dT}{dt} = k \quad \text{and} \quad \frac{1}{x} \frac{d^2 x}{dx^2} = k.$$

$$\frac{dT}{dt} = c^2 T k \quad \text{and} \quad \frac{d^2 x}{dx^2} = x k$$

$$\frac{dT}{dt} = c^2 T k = 0 \quad \text{and} \quad \frac{d^2 x}{dx^2} - x k = 0$$

$$(D - c^2 k) T = 0 \quad \text{and} \quad (D^2 - k) x = 0$$

$$\text{where } D = \frac{d}{dt} \quad \text{and} \quad D^2 = \frac{d^2}{dx^2}$$

The auxiliary Equations are

$$m - c^2 k = 0 \quad \text{and} \quad m^2 - k = 0$$

Case (i) :- Let $k = 0$

$$\text{Auxiliary Equation (A.E.) is } m = 0 \quad \text{and} \quad m^2 = 0$$

$m=0$, and $m=0,0$ are the roots.
Solutions are given by.

$$T = c_1 e^{0t} = c_1 \text{ and } x = c_2 e^{0x} + c_3 x e^{0x} \\ = (c_2 + c_3 x) e^{0x} \\ = (c_2 + c_3 x)$$

$$y = T \cdot x = c_1 (c_2 + c_3 x)$$

$$y = A + Bx \text{ where } A = c_1 c_2 \text{ and } B = c_1 c_3$$

Case (ii):- Let k be positive ($k = +p^2$)

$$\text{A.E.S:- } m - c^2 p^2 = 0 \text{ and } m^2 - p^2 = 0 \\ m = c^2 p^2 \text{ and } m = \pm p$$

The solution is given by

$$T = c_4 e^{c^2 p^2 t} \text{ and } x = c_5 e^{px} + c_6 e^{-px} \\ y = xT = c_4 e^{c^2 p^2 t} (c_5 e^{px} + c_6 e^{-px}) \\ = e^{c^2 p^2 t} (C e^{px} + D e^{-px})$$

Case (iii):- Let k be negative ($k = -p^2$)

$$\text{A.E.S} = m + c^2 p^2 = 0 \text{ and } m^2 + p^2 = 0 \\ m = -c^2 p^2 \text{ and } m^2 = -p^2 \\ m = \pm \sqrt{-p^2} \\ = \pm \sqrt{-1} p \\ m = \pm ip$$

The solution is given by

$$T = c_7 e^{-c^2 p^2 t} \text{ and } x = c_8 \cos(px) + c_9 \sin(px) \\ y = xT = c_7 e^{-c^2 p^2 t} (c_8 \cos(px) + c_9 \sin(px))$$

$$y = e^{-c^2 p^2 t} (E \cos(px) + F \sin(px)) \text{ where}$$

$$E = c_7 c_8 \text{ and } F = c_7 c_9$$

MODULE 4: LAPLACE TRANSFORM

PART-A

1) Show that $L[e^{at}] = \frac{1}{s-a}$

$$\rightarrow L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

$$= \int_0^{\infty} e^{-st} e^{at} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt \Rightarrow \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty}$$

$$= \frac{1}{(s-a)} [e^{-(s-a)t}]_0^{\infty}$$

$$= \frac{-1}{(s-a)} \left[\frac{1}{\infty} - 1 \right] \Rightarrow \frac{-1}{s-a} (0-1) \Rightarrow \frac{1}{s-a} //$$

2) State the division by "t" property of Laplace transform.

$\rightarrow$ If $L[f(t)] = F(s)$ then $L\left[\frac{f(t)}{t}\right] = \int_s^{\infty} F(s) ds$ provided integral exists

3) Find the inverse Laplace transform of $\frac{3s^2+4}{s^2}$

$\rightarrow$ We need to find $L^{-1}\left[\frac{3s^2+4}{s^2}\right]$

$$= L^{-1}\left[\frac{3}{s^3} + \frac{4}{s^2}\right]$$

By linearity property $L^{-1}\left[\frac{1}{s^n}\right] = \frac{t^{n-1}}{(n-1)!}$

$$= \frac{3 \cdot t^2}{2!} + \frac{4t^4}{4!} \Rightarrow \frac{3t^2}{2} + \frac{t^4}{6}$$

$$\Rightarrow \frac{3t^2}{2} + \frac{1}{6}t^4 //$$

4) Write the formula for shifting property of Heaviside shift theorem

$$\rightarrow L[f(t-a) \cdot u(t-a)] = e^{-as} F(s)$$

$$\text{where } L[f(t)] = F(s)$$

5) Write the formula for the Laplace transform of a periodic function.

$\rightarrow$ If $f(t)$ is a periodic function of period T then

$$L[f(t)] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

PART-B

1) Find the Laplace transform of $t^5 \cosh 3t$

$\rightarrow$ We know that $\cosh 3t = \frac{e^{3t} + e^{-3t}}{2}$

$$\text{Now } t^5 \cosh 3t = t^5 \left(\frac{e^{3t} + e^{-3t}}{2} \right)$$

$$= \frac{1}{2} (t^5 e^{3t} + t^5 e^{-3t})$$

We know the standard formula $L[t^n e^{at}] = \frac{n!}{(s-a)^{n+1}}$

$$L[t^5 \cosh 3t] = \frac{1}{2} (L[t^5 e^{3t}] + L[t^5 e^{-3t}])$$

$$= \frac{1}{2} \left(\frac{5!}{(s-3)^6} - \frac{5!}{(s+3)^6} \right)$$

$$= \frac{1}{2} \left(\frac{120}{(s-3)^6} + \frac{120}{(s+3)^6} \right)$$

$$= 60 \left(\frac{1}{(s-3)^6} + \frac{1}{(s+3)^6} \right)$$

2.) Express $f(t) = \begin{cases} t^2, & 0 < t \leq 2 \\ 4t, & t > 2 \end{cases}$ in terms of unit step function

and hence evaluate it.

→ By property: $f(t) = f(t) + [f_2(t) - f_1(t)] u(t-a)$
 $= t^2 + [4t - t^2] u(t-2)$

By applying Laplace transform:

$$L[f(t)] = L[t^2] + L[(4t - t^2) u(t-2)] \quad (\because a=2)$$

Taking $F(t-a) = 4t - t^2 \Rightarrow F(t-2) = 4t - t^2$ [Replace t w $t+2$]

$$F(t+2-2) = 4(t+2) - (t+2)^2 \Rightarrow F(t) = 4t + 8 - (t^2 + 4t + 4)$$

$$= F(t) = 8 - t^2 - 4 \Rightarrow F(t) = 4 - t^2 \Rightarrow L[F(t)] = L[4 - t^2]$$

$$= L[4] - L[t^2] \Rightarrow F(s) = \frac{4}{s} - \frac{2}{s^3}$$

$$L[F(t-2) u(t-2)] = e^{-2s} F(s) \Rightarrow L[(4t - t^2) u(t-2)] = e^{-2s} \left(\frac{4}{s} - \frac{2}{s^3} \right)$$

Then eqn (1) becomes $L[f(t)] = \frac{2}{s^3} + e^{-2s} \left(\frac{4}{s} - \frac{2}{s^3} \right)$

3.) Find the Laplace transform of the periodic function $f(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4 \end{cases}$

→ Period $T = 4 - 0 = 4$

For a periodic function, the Laplace transform formula is,

$$L[f(t)] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$= \frac{1}{1 - e^{-4s}} \int_0^4 e^{-st} f(t) dt$$

$$= \frac{1}{1 - e^{-4s}} \left[\int_0^2 e^{-st} 3t dt + \int_2^4 e^{-st} 6 dt \right]$$

First integral: $3 \int_0^2 t e^{-st} dt$

$$= 3 \left\{ \left[t \frac{e^{-st}}{s} \right]_0^2 - \int_0^2 1 \cdot \frac{e^{-st}}{-s} dt \right\}$$

$$= 3 \left\{ \left[\frac{t e^{-st}}{-s} \right]_0^2 - \left[\frac{e^{-st}}{s^2} \right]_0^2 \right\}$$

$$= 3 \left\{ \left[\frac{2 e^{-2s}}{-s} - 0 \right] - \left[\frac{e^{-2s}}{s^2} - \frac{1}{s^2} \right] \right\}$$

$$= 3 \left\{ -\frac{2s e^{-2s}}{s^2} - \frac{e^{-2s}}{s^2} + \frac{1}{s^2} \right\}$$

$$= \frac{3}{s^2} [1 - (1+2s) e^{-2s}]$$

Second integral: $6 \int_2^4 e^{-st} dt = 6 \left[\frac{e^{-st}}{-s} \right]_2^4$

$$= 6 \left[\frac{e^{-4s}}{-s} - \frac{e^{-2s}}{-s} \right]$$

$$= 6 \left[-\frac{e^{-4s}}{s} + \frac{e^{-2s}}{s} \right]$$

$$= \frac{6}{s} [e^{-2s} - e^{-4s}]$$

$$L[f(t)] = \frac{1}{1 - e^{-4s}} \left[\frac{3}{s^2} (1 - (1+2s) e^{-2s}) + \frac{6}{s} [e^{-2s} - e^{-4s}] \right]$$

4) Evaluate using Laplace transform $\int_0^\infty t e^{-2t} \sin t dt$

→ Given: $\int_0^\infty t e^{-2t} \sin t dt$

$$\int_0^\infty e^{-2t} (t \sin t) dt$$

Compare the above integral with the Laplace transform

formula $L[f(t)] = \int_0^\infty e^{-st} f(t) dt$

Here $f(t) = t \sin t$, $s = 2$

$$L[t(\sin t)] = (-1) \cdot \frac{d}{ds} \left[\frac{1}{s^2+1} \right]$$

$$= \frac{-d}{ds} [s^2+1]^{-1}$$

$$= (-1)(-1)(s^2+1)^{-2} \times 2s$$

$$L[t(\sin t)] = \frac{2s}{(s^2+1)^2}$$

$$\therefore \int_0^{\infty} e^{-2t} t \sin t dt = \frac{2 \times 2}{(2^2+1)^2}$$

$$\int_0^{\infty} e^{-2t} t \sin t dt = \frac{4}{25}$$

PART-C

1. Find $L\left[\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3\right]$

$$\rightarrow \text{Given: } \left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3 = t^{3/2} + 3t \times \frac{1}{t^{1/2}} + 3t^{1/2} \frac{1}{t} + \frac{1}{t^{3/2}}$$

$$= t^{3/2} + 3t^{1/2} + 3t^{-1/2} + t^{-3/2}$$

$$L\left[\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3\right] = L(t^{3/2}) + 3L[t^{1/2}] + 3L[t^{-1/2}] + L[t^{-3/2}]$$

$$\therefore L(t^n) = \frac{\Gamma(n+1)}{s^{n+1}}$$

$$\therefore L\left[\left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^3\right] = \frac{\Gamma(5/2)}{s^{5/2}} + 3 \frac{\Gamma(3/2)}{s^{3/2}} + 3 \frac{\Gamma(1/2)}{s^{1/2}} + \frac{\Gamma(-1/2)}{s^{-1/2}}$$

$$= \frac{3}{4} \frac{\sqrt{\pi}}{s^{5/2}} + 3 \left[\frac{1}{2} \frac{\sqrt{\pi}}{s^{3/2}} \right] + 3 \frac{\sqrt{\pi}}{s^{1/2}} - \frac{2\sqrt{\pi}}{s^{-1/2}}$$

[5]

$$= \frac{3}{4} \frac{\sqrt{\pi}}{s^{5/2}} + \frac{3}{2} \frac{\sqrt{\pi}}{s^{3/2}} + 3 \frac{\sqrt{\pi}}{s^{1/2}} - \frac{2\sqrt{\pi}}{s^{-1/2}}$$

$$= \frac{\sqrt{\pi}}{4} \left[\frac{3}{s^{5/2}} + \frac{6}{s^{3/2}} + \frac{12}{s^{1/2}} - \frac{8}{s^{-1/2}} \right]$$

2. Find the inverse Laplace transform of $\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6}$

$$\rightarrow \text{Given : } L^{-1} \left[\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6} \right]$$

$$= \frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6}$$

Factorise the denominator

$$\frac{2s^2 - 6s + 5}{(s-1)(s-2)(s-3)}$$

By partial fraction decomposition

$$\frac{2s^2 - 6s + 5}{(s-1)(s-2)(s-3)} = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-3)} \quad \text{--- (1)}$$

By multiplying whole eqn by $(s-1)(s-2)(s-3)$

$$2s^2 - 6s + 5 = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

$$\text{Put } s=1$$

$$\Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$$

$$\text{Put } s=2 : 1 = -B$$

$$B = -1$$

$$\text{Put } s=3 : 5 = 2C$$

$$C = \frac{5}{2}$$

Then eqn (1) becomes:

$$\frac{2s^2 - 6s + 5}{(s-1)(s-2)(s-3)} = \frac{1/2}{(s-1)} + \frac{(-1)}{(s-2)} + \frac{(5/2)}{(s-3)}$$

$$= \frac{1}{2(s-1)} - \frac{1}{(s-2)} + \frac{5}{2(s-3)}$$

Apply inverse Laplace transform on both sides

$$L^{-1} \left[\frac{2s^2 - 6s + 5}{(s-1)(s-2)(s-3)} \right] = L^{-1} \left(\frac{1}{2(s-1)} \right) - L^{-1} \left(\frac{1}{(s-2)} \right) + L^{-1} \left(\frac{5}{2(s-3)} \right)$$

$$= L^{-1} \left[\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6} \right] = \frac{1}{2} e^t - e^{2t} + \frac{5}{2} e^{3t}$$

3. Solve by Laplace transform method $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$

→ Taking Laplace transform on both sides of the given equation we have,

$$L[y''(t)] + 4L[y'(t)] + 3L[y(t)] = L[e^{-t}]$$

$$[s^2 L[y(t)] - s y(0) - y'(0)] + 4[s L[y(t)] - y(0)] + 3L[y(t)] = \frac{1}{s+1}$$

Using the given initial conditions we obtain.

$$(s^2 + 4s + 3) L[y(t)] - s - 1 - 4 = \frac{1}{s+1}$$

$$(s^2 + 4s + 3) L[y(t)] = (s+5) + \frac{1}{(s+1)}$$

$$(s+1)(s+3) L[y(t)] = \frac{s^2 + 6s + 6}{s+1}$$

$$L[y(t)] = \frac{s^2 + 6s + 6}{(s+1)^2(s+3)}$$

$$y(t) = L^{-1} \left[\frac{s^2 + 6s + 6}{(s+1)^2(s+3)} \right]$$

$$\text{Let } \frac{s^2 + 6s + 6}{(s+1)^2(s+3)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{s+3}$$

Multiplying by $(s+1)^2(s+3)$ we get,

$$s^2 + 6s + 6 = A(s+1)(s+3) + B(s+3) + C(s+1)^2 \quad \text{--- (1)}$$

$$\text{Put } s = -1 : 1 = 2B \Rightarrow B = 1/2$$

$$\text{Put } s = -3 : -3 = 4C \Rightarrow C = -3/4$$

Equating the coefficient of s^2 on both sides of (1) we get,

$$1 = A + C \Rightarrow A = 7/4$$

$$L^{-1} \left[\frac{s^2 + 6s + 6}{(s+1)^2(s+3)} \right] = \frac{7}{4} L^{-1} \left[\frac{1}{s+1} \right] + \frac{1}{2} L^{-1} \left[\frac{1}{(s+1)^2} \right] - \frac{3}{4} L^{-1} \left[\frac{1}{s+3} \right]$$

$$y(t) = \frac{7}{4} e^{-t} + \frac{1}{2} e^{-t} - \frac{3}{4} e^{-3t}$$